

Synthesis Review: Vector Motion, Polar Curves, and Accumulation

Vector motion, polar curve intersections, and accumulated quantities

Directions

- Problems 1–3 are a warm-up, one on each idea. After that they are mixed: decide what is being asked before you pick a formula.
- Angles are in radians. Give exact answers where the problem says so, and decimals to the stated precision otherwise.
- Where an answer carries a unit, the answer line names it.

1. A particle moves in the plane with position $x(t) = t^2 + 1$ and $y(t) = 3t$. Find its speed at $t = 2$.

Answer: _____

2. Find every angle θ in $[0, 2\pi)$ at which the polar curves $r = 2$ and $r = 4 \cos \theta$ meet. Give exact answers.

Answer: _____

3. Water enters a reservoir at the rate $r(t) = 5 + 2 \cos t$ litres per hour. How much water enters between $t = 0$ and $t = \pi$ hours? Give an exact answer and a decimal to three places.

Answer: _____ L

4. A curve is given by $x = \sin(2t)$, $y = t^2$.

(a) Find $\frac{dx}{dt}$.

Answer: _____

(b) Find $\frac{dy}{dx}$ at $t = \frac{\pi}{6}$, to three decimal places.

Answer: _____

5. Find the area of the region that lies inside the polar curve $r = 4 \cos \theta$ and outside the circle $r = 2$. Give an exact answer and a decimal to three places.

Answer: _____

6. A particle travels along the curve $x(t) = 3t$, $y(t) = 2t^{3/2}$ for $0 \leq t \leq 4$, with lengths in metres. Find the distance it travels. Give an exact answer and a decimal to three places.

Answer: _____ m

7. A particle in the plane has velocity components $x'(t) = 4 \cos t$ and $y'(t) = 4 \sin t$, and at $t = 0$ it is at the point $(2, 0)$. Find its position at $t = \frac{\pi}{2}$.

Answer: _____

8. Consider the polar curves $r = 1 + \cos \theta$ and $r = 3 \cos \theta$.

(a) Find every θ in $[0, 2\pi)$ satisfying both equations.

Answer: _____

(b) Find the value of r at those angles.

Answer: _____

(c) The two curves also both pass through the pole. Explain why solving the two equations simultaneously does not find that point.

9. Water flows through a pipe at the rate $r(t) = 12e^{-0.2t}$ litres per minute, where t is in minutes.

(a) How much water flows through during the first 5 minutes? Give an exact answer and a decimal to two places.

Answer: _____ L

(b) At what time has the flow rate fallen to half its initial value? Give a decimal to three places.

Answer: _____ min

10. A particle starts at the origin and moves with velocity vector $\langle 2t, t^2 - 1 \rangle$ metres per second, for $0 \leq t \leq 3$ seconds.

(a) Find the total distance the particle travels on $[0, 3]$.

Answer: _____ m

(b) Find the particle's position at $t = 3$.

Answer: _____

(c) Explain why the distance in part (a) is greater than the straight-line distance from the origin to the position in part (b).